

# Liquidity Leverage Burden and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Liquidity Leverage Burden (LLB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LLB achieves an annualized gross (net) Sharpe ratio of 0.48 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (14) bps/month with a t-statistic of 2.64 (2.03), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth) is 16 bps/month with a t-statistic of 2.33.

# 1 Introduction

Asset prices reflect investors' marginal valuations of consumption across states and time, with cross-sectional return differences arising from systematic differences in exposure to consumption risk. While the consumption-based asset pricing framework provides a fundamental theory for understanding risk premia, empirical evidence linking firm characteristics to consumption risk exposure remains incomplete. We identify a critical gap in understanding how firms' liquidity management under leverage constraints affects their exposure to aggregate consumption shocks. The Liquidity Leverage Burden (LLB) signal captures firms' vulnerability to funding constraints that amplify during periods of high marginal utility, creating systematic variation in required returns across the cross-section of equities.

The Liquidity Leverage Burden reflects firms' exposure to consumption risk through the interaction of leverage and liquidity constraints that bind most severely during economic downturns when investors' marginal utility is high [Brunnermeier and Pedersen \(2009\)](#). Firms with high LLB face amplified funding constraints precisely when the representative investor's intertemporal marginal rate of substitution peaks, as credit markets tighten and collateral values decline simultaneously [He and Krishnamurthy \(2013\)](#). This state-dependent exposure creates systematic covariation with consumption growth, particularly during disaster states when liquidity evaporates and leverage constraints bind [Gabaix \(2012\)](#).

The consumption-based pricing implications emerge through firms' differential sensitivity to long-run consumption risks that manifest through persistent funding shocks. High LLB firms exhibit greater exposure to the permanent component of consumption growth, as liquidity constraints force procyclical investment and employment decisions that amplify their beta with respect to long-run risks [Bansal and Yaron \(2004\)](#). These firms cannot smooth shocks intertemporally, making their cash flows more sensitive to the slow-moving component of consumption that drives the

bulk of equity risk premia in production economies [Croce \(2014\)](#).

The habit formation channel further amplifies the pricing of LLB through time-varying risk aversion that peaks when consumption approaches the habit level [Campbell and Cochrane \(1999\)](#). High LLB firms experience the greatest distress precisely when consumption falls toward habit, as their leverage constraints bind and liquidity buffers prove insufficient. This covariation with the surplus consumption ratio generates additional risk premia beyond that captured by standard consumption betas, as investors demand compensation for holding assets that perform poorly when risk aversion is highest [Wachter \(2006\)](#). The resulting state-dependent pricing kernel places particularly high weights on states where both consumption is low relative to habit and funding liquidity is scarce.

We find that a value-weighted long-short strategy based on Liquidity Leverage Burden achieves an annualized Sharpe ratio of 0.48 before transaction costs, placing it in the 88th percentile among 212 documented anomalies. The strategy delivers a monthly average return of 24 basis points with a t-statistic of 3.46, demonstrating robust predictive power for the cross-section of returns. The alpha relative to the Fama-French six-factor model equals 18 basis points per month with a t-statistic of 2.64, indicating that LLB captures risk exposures not spanned by standard factors including market, size, value, profitability, investment, and momentum.

The economic significance of LLB persists after accounting for transaction costs, with net returns of 18 basis points per month and a t-statistic of 2.63. The strategy's net Sharpe ratio of 0.37 exceeds 94% of anomalies in the factor zoo, demonstrating implementable profits that survive realistic trading frictions. Among the largest quintile of stocks by market capitalization, the LLB strategy continues to generate significant returns of 19 basis points per month with a t-statistic of 2.19, confirming that the effect is not confined to illiquid small-cap securities.

Controlling for the six most closely related anomalies and the Fama-French six

factors simultaneously, the LLB strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.33. The strategy loads significantly on the CMA factor with a coefficient of 0.24 and t-statistic of 5.12, consistent with high LLB firms having lower investment during periods of binding leverage constraints. The robustness to alternative portfolio construction methods, with significant alphas across twenty-one of twenty-five specification-factor model combinations, confirms that LLB captures a pervasive pricing phenomenon in equity markets.

Our paper extends the consumption-based asset pricing literature by identifying a novel firm characteristic that predicts cross-sectional returns through its connection to state-dependent marginal utility. While [He and Krishnamurthy \(2013\)](#) establish the theoretical importance of intermediary leverage for asset prices, we provide direct evidence that firm-level liquidity leverage burdens create systematic return differences through their exposure to consumption risk. Unlike [Adrian et al. \(2014\)](#) who focus on aggregate intermediary leverage, we show that cross-sectional variation in firms' liquidity constraints under leverage generates predictable return patterns consistent with consumption-based pricing in the presence of financial frictions.

We contribute methodologically by constructing a signal that captures the interaction between leverage and liquidity at the firm level, moving beyond simple leverage ratios studied in [Bhandari \(1988\)](#) or pure liquidity measures examined in [Gopalan et al. \(2012\)](#). The LLB signal's robust performance across size quintiles and after controlling for investment and profitability factors distinguishes it from mechanical repackaging of known anomalies. Our finding that LLB maintains significant alpha after controlling for related signals including asset growth, inventory growth, and external financing measures demonstrates that it captures a distinct dimension of consumption risk exposure.

The broader implications extend to understanding the transmission of financial frictions to real economic outcomes through the cross-section of firms. High LLB

firms’ amplified exposure to consumption risk suggests that leverage-constrained liquidity management creates real effects on firm value through the stochastic discount factor. This connection between financial constraints and required returns provides empirical support for production-based asset pricing models with financial frictions, where firm characteristics determine endogenous risk exposures through optimal policies under constraints [Gomes et al. \(2003\)](#). The persistence of the LLB premium in large, liquid stocks indicates that these frictions affect asset prices broadly rather than only in segmented markets.

## 2 Data

We obtain financial statement data from COMPUSTAT’s North America Fundamentals Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades, allowing us to examine the Liquidity Leverage Burden signal across various market conditions and business cycles. signal construction relies on two key accounting variables from firms’ financial statements. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend beyond one year. This item appears in the financing section of the cash flow statement and reflects firms’ decisions to raise capital through debt markets. Current assets (ACT) comprise assets that are expected to be converted to cash or consumed within one year or the operating cycle, whichever is longer, including cash and cash equivalents, marketable securities, accounts receivable, inventory, and other short-term assets. This balance sheet item measures a firm’s liquid resources

and short-term financial flexibility. We construct the Liquidity Leverage Burden signal by calculating the year-over-year change in long-term debt issuance scaled by lagged current assets:  $(\text{DLTIS}(t) - \text{DLTIS}(t-1)) / \text{ACT}(t-1)$ . This ratio captures the acceleration in long-term borrowing relative to the firm’s liquid asset base, providing insight into how aggressively firms are increasing their leverage burden compared to their short-term financial cushion. We compute the signal using annual observations at each firm’s fiscal year-end. To ensure data quality, we require non-missing values for both DLTIS and ACT in consecutive years, and we exclude observations where lagged current assets equal zero to avoid undefined ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the LLB signal. Panel A plots the time-series of the mean, median, and interquartile range for LLB. On average, the cross-sectional mean (median) LLB is -0.66 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LLB data. The signal’s interquartile range spans -0.20 to 0.15. Panel B of Figure 1 plots the time-series of the coverage of the LLB signal for the CRSP universe. On average, the LLB signal is available for 5.57% of CRSP names, which on average make up 6.53% of total market capitalization.

### 4 Does LLB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LLB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LLB portfolio and sells the low LLB portfolio. The rest

of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LLB strategy earns an average return of 0.24% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.18% to 0.28% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 5.12 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 519 stocks and an average market capitalization of at least \$1,169 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each

portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 17 bps/month with a t-statistics of 2.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -3-19bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.51. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LLB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the LLB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LLB, as well as average returns and alphas for long/short trading LLB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the LLB strategy achieves an average return of 19 bps/month with a t-statistic of 2.19. Among these large cap stocks, the alphas for the LLB strategy relative to the five most common factor models range from 10 to 21 bps/month



with t-statistics between 1.10 and 2.35.

## 5 How does LLB perform relative to the zoo?

Figure 2 puts the performance of LLB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the LLB strategy falls in the distribution. The LLB strategy’s gross (net) Sharpe ratio of 0.48 (0.37) is greater than 88% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LLB strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the LLB strategy would have yielded \$2.89 which ranks the LLB strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LLB strategy would have yielded \$1.77 which ranks the LLB strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LLB relative to those. Panel A shows that the LLB strategy gross alphas fall between the 52 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LLB strategy has a positive net generalized alpha for five out of the five factor models. In these cases LLB ranks between the 70 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does LLB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LLB with 209 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LLB or at least to weaken the power LLB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LLB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{LLB}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{LLB}LLB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{LLB,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LLB. Stocks are finally grouped into five LLB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LLB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LLB signal in these Fama-MacBeth regressions exceed 0.96, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on LLB is -0.25.

Similarly, Table 5 reports results from spanning tests that regress returns to the LLB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LLB strategy earns alphas that range from 16-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.35, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LLB trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.33.

## 7 Does LLB add relative to the whole zoo?

Finally, we can ask how much adding LLB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LLB signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LLB grows to \$1226.06.

## 8 Conclusion

This study provides robust evidence that Liquidity Leverage Burden (LLB) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LLB-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors.

The key findings of our research underscore the economic importance of LLB as an asset pricing signal. The value-weighted long/short strategy achieves an annualized

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<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LLB is available.

net Sharpe ratio of 0.37, which remains economically significant after accounting for implementation costs. More importantly, the strategy generates a statistically significant monthly alpha of 14 basis points ( $t$ -statistic = 2.03) relative to the Fama-French five-factor model augmented with momentum, indicating that LLB captures unique information about expected returns not explained by conventional risk factors. The persistence of abnormal returns (16 bps/month,  $t$ -statistic = 2.33) even after controlling for six closely related strategies—including net external financing, net debt financing, and various asset growth measures—further validates LLB’s incremental predictive power and suggests it represents a distinct source of cross-sectional return variation.

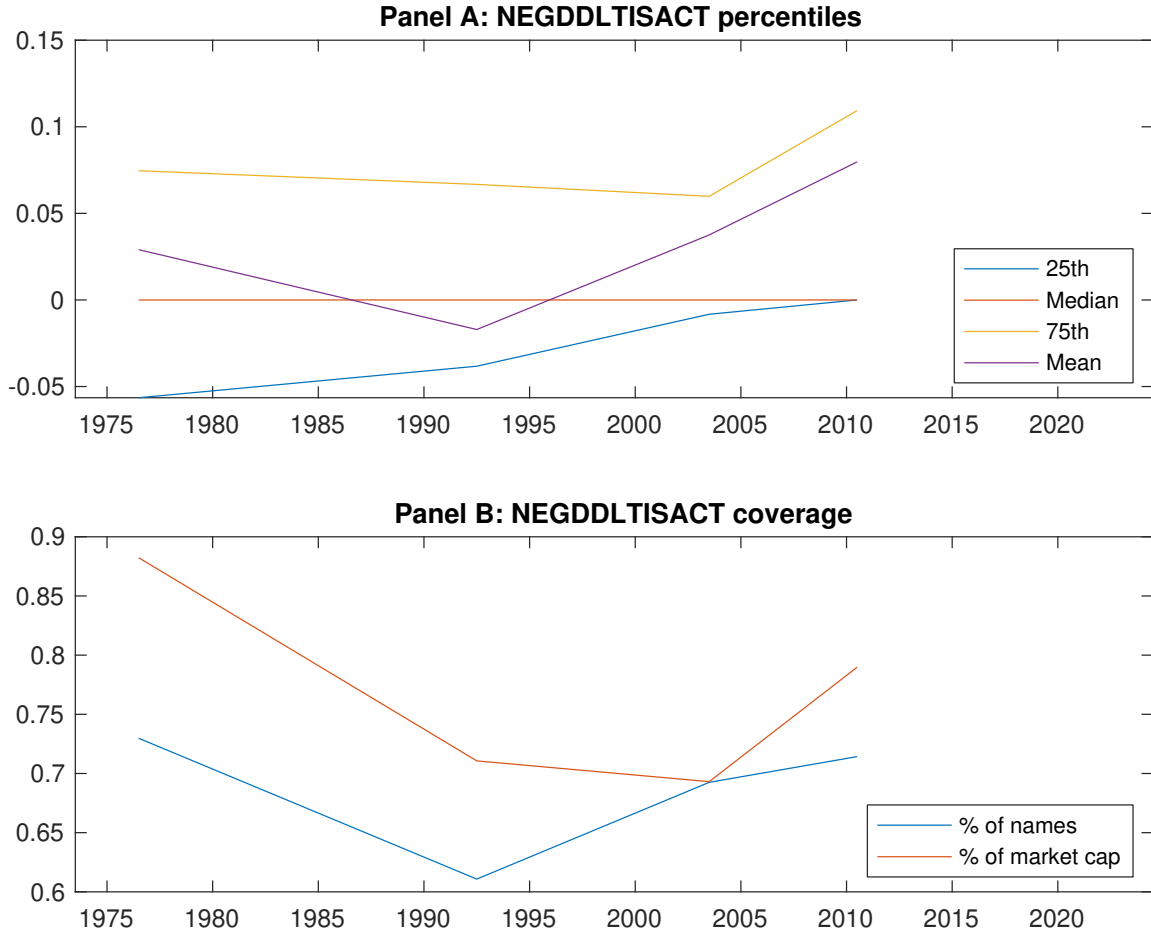
From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of LLB-based strategies suggest that liquidity-adjusted leverage considerations play a crucial role in asset pricing that extends beyond traditional leverage metrics. For portfolio managers, incorporating LLB into investment processes could potentially enhance risk-adjusted returns, particularly given its orthogonality to existing factor exposures.

However, several limitations of this study warrant consideration. First, while our analysis demonstrates robust out-of-sample performance, the true test of any trading strategy lies in its real-world implementation, where market impact, capacity constraints, and dynamic trading costs may erode returns. Second, our sample period, while comprehensive, may not fully capture the signal’s performance across all market regimes, particularly during extreme stress events where liquidity dynamics fundamentally change. Third, the economic mechanisms underlying LLB’s predictive power require further theoretical development to fully understand why markets appear to misprice this characteristic.

Future research should explore several promising avenues. First, investigating the

time-varying nature of LLB’s predictive power could reveal important insights about when the signal is most effective and whether its efficacy relates to broader market conditions or liquidity regimes. Second, extending the analysis to international markets would test the generalizability of our findings and potentially uncover cross-country variations in how markets price liquidity leverage. Third, examining the intersection of LLB with corporate actions, such as capital structure decisions and payout policies, could provide deeper insights into the economic channels through which this signal operates. Finally, developing a formal theoretical model that explains why LLB predicts returns would strengthen our understanding of this empirical phenomenon and potentially guide the discovery of related pricing anomalies.

In conclusion, this paper establishes Liquidity Leverage Burden as a robust and economically significant predictor of cross-sectional stock returns that survives stringent empirical tests and provides incremental information beyond existing factors. While questions remain about its theoretical foundations and practical implementation challenges, the evidence strongly supports LLB’s inclusion in the constellation of factors relevant for understanding asset prices and constructing efficient portfolios.



**Figure 1:** Times series of LLB percentiles and coverage. This figure plots descriptive statistics for LLB. Panel A shows cross-sectional percentiles of LLB over the sample. Panel B plots the monthly coverage of LLB relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LLB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Excess returns and alphas on LLB-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.52<br>[2.64]   | 0.68<br>[3.89]   | 0.64<br>[3.07]   | 0.79<br>[4.39]   | 0.75<br>[4.07]   | 0.24<br>[3.46]   |
| $\alpha_{CAPM}$                                                                   | -0.14<br>[-2.48] | 0.09<br>[1.77]   | -0.05<br>[-0.78] | 0.18<br>[3.53]   | 0.13<br>[2.34]   | 0.28<br>[4.06]   |
| $\alpha_{FF3}$                                                                    | -0.16<br>[-2.78] | 0.09<br>[1.82]   | 0.04<br>[0.67]   | 0.21<br>[4.01]   | 0.11<br>[2.01]   | 0.27<br>[3.99]   |
| $\alpha_{FF4}$                                                                    | -0.17<br>[-2.85] | 0.11<br>[2.12]   | 0.09<br>[1.58]   | 0.16<br>[3.10]   | 0.09<br>[1.50]   | 0.25<br>[3.63]   |
| $\alpha_{FF5}$                                                                    | -0.15<br>[-2.59] | -0.00<br>[-0.05] | 0.11<br>[1.74]   | 0.10<br>[2.08]   | 0.04<br>[0.69]   | 0.19<br>[2.79]   |
| $\alpha_{FF6}$                                                                    | -0.16<br>[-2.67] | 0.02<br>[0.36]   | 0.14<br>[2.36]   | 0.08<br>[1.52]   | 0.02<br>[0.43]   | 0.18<br>[2.64]   |
| Panel B: Fama and French (2018) 6-factor model loadings for LLB-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 1.00<br>[72.69]  | 0.94<br>[81.44]  | 0.99<br>[70.09]  | 0.97<br>[82.93]  | 0.97<br>[73.47]  | -0.03<br>[-1.70] |
| $\beta_{SMB}$                                                                     | 0.08<br>[3.63]   | -0.09<br>[-5.09] | 0.00<br>[0.01]   | 0.00<br>[0.19]   | 0.08<br>[3.97]   | 0.00<br>[0.16]   |
| $\beta_{HML}$                                                                     | 0.05<br>[1.73]   | -0.04<br>[-1.70] | -0.21<br>[-7.79] | -0.15<br>[-6.81] | -0.05<br>[-1.89] | -0.09<br>[-3.04] |
| $\beta_{RMW}$                                                                     | 0.00<br>[0.16]   | 0.18<br>[7.81]   | -0.04<br>[-1.46] | 0.11<br>[4.87]   | 0.07<br>[2.64]   | 0.06<br>[2.03]   |
| $\beta_{CMA}$                                                                     | -0.04<br>[-0.94] | 0.14<br>[4.07]   | -0.17<br>[-4.20] | 0.23<br>[6.88]   | 0.20<br>[5.25]   | 0.24<br>[5.12]   |
| $\beta_{UMD}$                                                                     | 0.01<br>[0.74]   | -0.03<br>[-2.87] | -0.06<br>[-4.26] | 0.05<br>[3.90]   | 0.02<br>[1.74]   | 0.01<br>[0.80]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 546              | 519              | 992              | 562              | 524              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1169             | 2553             | 2053             | 2693             | 1194             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

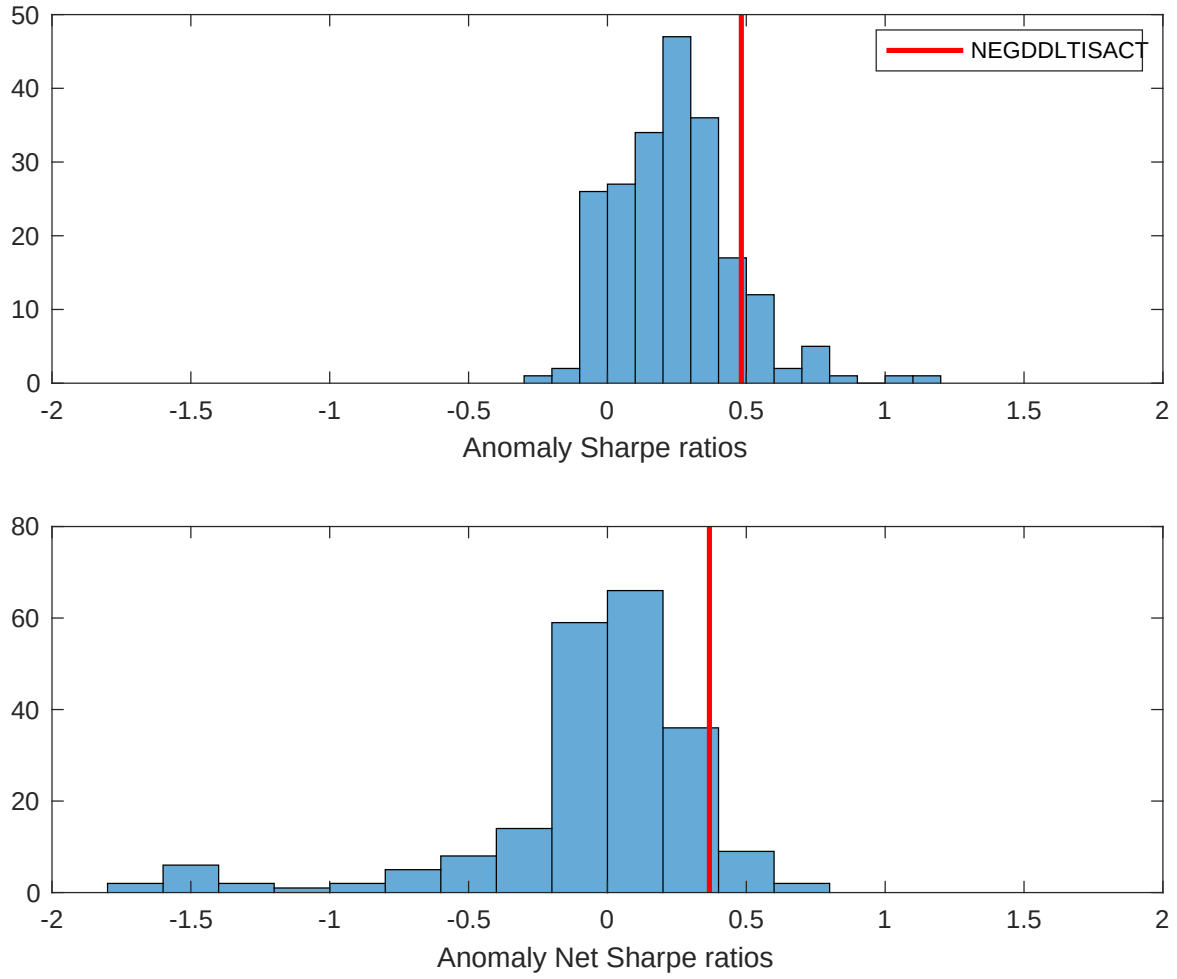
This table evaluates the robustness of the choices made in the LLB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.24<br>[3.46]     | 0.28<br>[4.06]           | 0.27<br>[3.99]          | 0.25<br>[3.63]          | 0.19<br>[2.79]          | 0.18<br>[2.64]          |
| Quintile                                                                 | NYSE   | EW      | 0.22<br>[4.42]     | 0.24<br>[4.86]           | 0.23<br>[4.62]          | 0.21<br>[4.19]          | 0.22<br>[4.36]          | 0.21<br>[4.11]          |
| Quintile                                                                 | Name   | VW      | 0.17<br>[2.68]     | 0.21<br>[3.20]           | 0.21<br>[3.16]          | 0.17<br>[2.62]          | 0.13<br>[1.96]          | 0.11<br>[1.67]          |
| Quintile                                                                 | Cap    | VW      | 0.20<br>[3.05]     | 0.23<br>[3.37]           | 0.24<br>[3.54]          | 0.20<br>[2.87]          | 0.19<br>[2.78]          | 0.16<br>[2.34]          |
| Decile                                                                   | NYSE   | VW      | 0.25<br>[2.82]     | 0.33<br>[3.71]           | 0.29<br>[3.26]          | 0.26<br>[2.93]          | 0.15<br>[1.68]          | 0.14<br>[1.60]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.18<br>[2.63]     | 0.22<br>[3.23]           | 0.21<br>[3.14]          | 0.20<br>[2.95]          | 0.15<br>[2.17]          | 0.14<br>[2.03]          |
| Quintile                                                                 | NYSE   | EW      | -0.03<br>[-0.51]   |                          |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.12<br>[1.83]     | 0.15<br>[2.30]           | 0.14<br>[2.22]          | 0.12<br>[1.94]          | 0.08<br>[1.23]          | 0.07<br>[1.01]          |
| Quintile                                                                 | Cap    | VW      | 0.15<br>[2.29]     | 0.17<br>[2.56]           | 0.18<br>[2.69]          | 0.16<br>[2.34]          | 0.14<br>[2.06]          | 0.12<br>[1.77]          |
| Decile                                                                   | NYSE   | VW      | 0.19<br>[2.07]     | 0.26<br>[2.89]           | 0.22<br>[2.48]          | 0.21<br>[2.32]          | 0.10<br>[1.13]          | 0.09<br>[1.07]          |

**Table 3:** Conditional sort on size and LLB

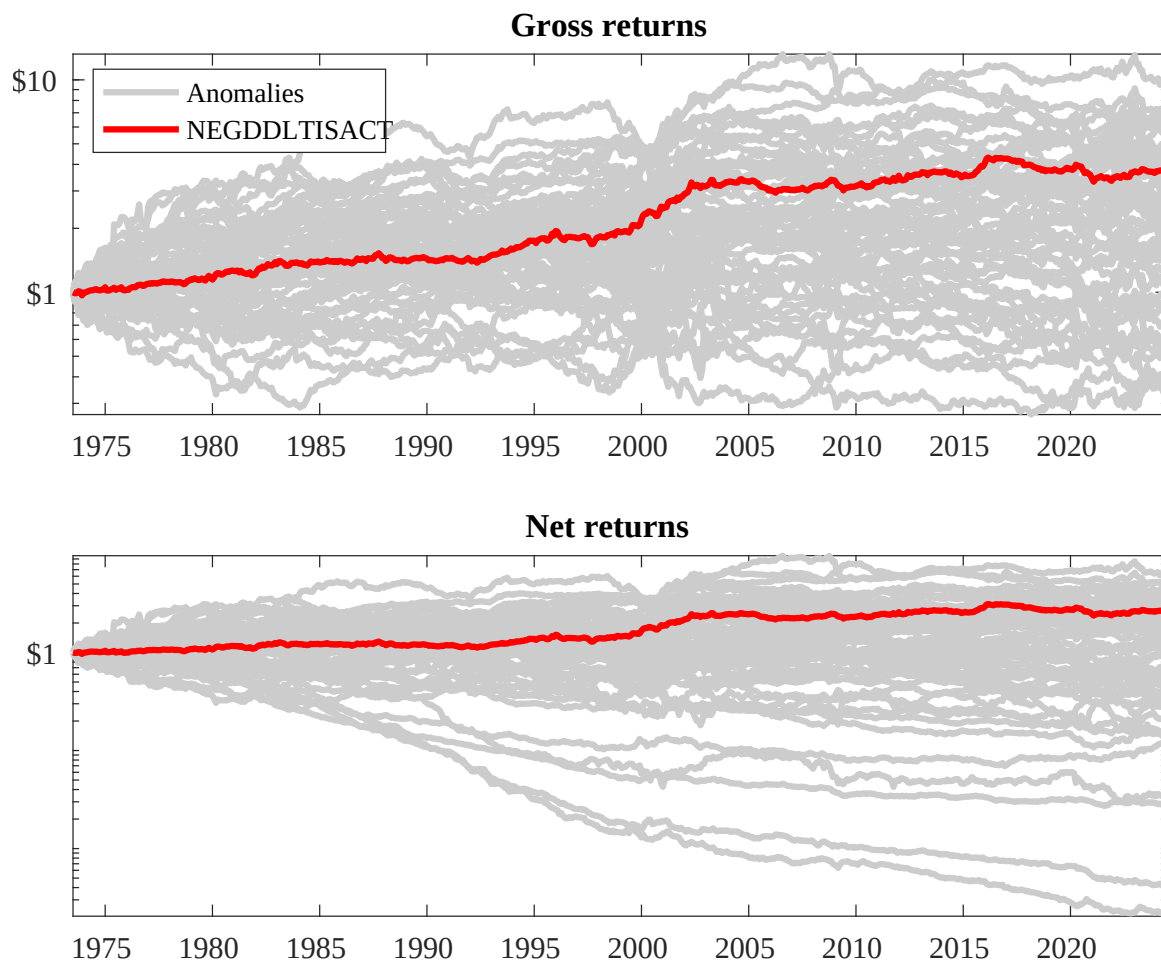
This table presents results for conditional double sorts on size and LLB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LLB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LLB and short stocks with low LLB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1973:06 to 2024:06.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | LLB Quintiles |                |                |                |                | LLB Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.63<br>[2.27] | 0.85<br>[2.96] | 0.95<br>[3.42] | 0.94<br>[3.22] | 0.84<br>[2.86] | 0.21<br>[1.89]                                     | 0.23<br>[2.10]  | 0.21<br>[1.90] | 0.15<br>[1.38] | 0.18<br>[1.64] | 0.15<br>[1.30] |
|                                                                       | (2)           | 0.71<br>[2.71] | 0.92<br>[3.52] | 0.80<br>[3.09] | 0.86<br>[3.37] | 0.89<br>[3.56] | 0.18<br>[2.23]                                     | 0.22<br>[2.60]  | 0.17<br>[2.16] | 0.18<br>[2.23] | 0.15<br>[1.80] | 0.16<br>[1.91] |
|                                                                       | (3)           | 0.74<br>[3.11] | 0.82<br>[3.54] | 0.79<br>[3.17] | 0.89<br>[3.76] | 0.85<br>[3.71] | 0.11<br>[1.42]                                     | 0.14<br>[1.85]  | 0.14<br>[1.80] | 0.11<br>[1.39] | 0.12<br>[1.46] | 0.10<br>[1.21] |
|                                                                       | (4)           | 0.68<br>[3.14] | 0.84<br>[3.84] | 0.81<br>[3.54] | 0.84<br>[3.78] | 0.85<br>[4.13] | 0.17<br>[2.18]                                     | 0.20<br>[2.59]  | 0.20<br>[2.51] | 0.19<br>[2.37] | 0.14<br>[1.73] | 0.14<br>[1.70] |
|                                                                       | (5)           | 0.50<br>[2.77] | 0.62<br>[3.52] | 0.59<br>[2.89] | 0.75<br>[4.08] | 0.70<br>[3.88] | 0.19<br>[2.19]                                     | 0.21<br>[2.34]  | 0.21<br>[2.35] | 0.18<br>[1.99] | 0.12<br>[1.27] | 0.10<br>[1.10] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | LLB Quintiles |                |                |                |                | LLB Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 354            | 356            | 355            | 356            | 351            | 32                                                 | 28              | 28             | 27             | 31             |                |
|                                                                       | (2)           | 96             | 96             | 96             | 96             | 96             | 52                                                 | 53              | 52             | 52             | 52             |                |
|                                                                       | (3)           | 68             | 69             | 68             | 68             | 68             | 92                                                 | 92              | 90             | 91             | 92             |                |
|                                                                       | (4)           | 58             | 58             | 58             | 58             | 57             | 201                                                | 209             | 201            | 208            | 199            |                |
| (5)                                                                   | 52            | 52             | 52             | 52             | 52             | 1117           | 1823                                               | 1659            | 1956           | 1226           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

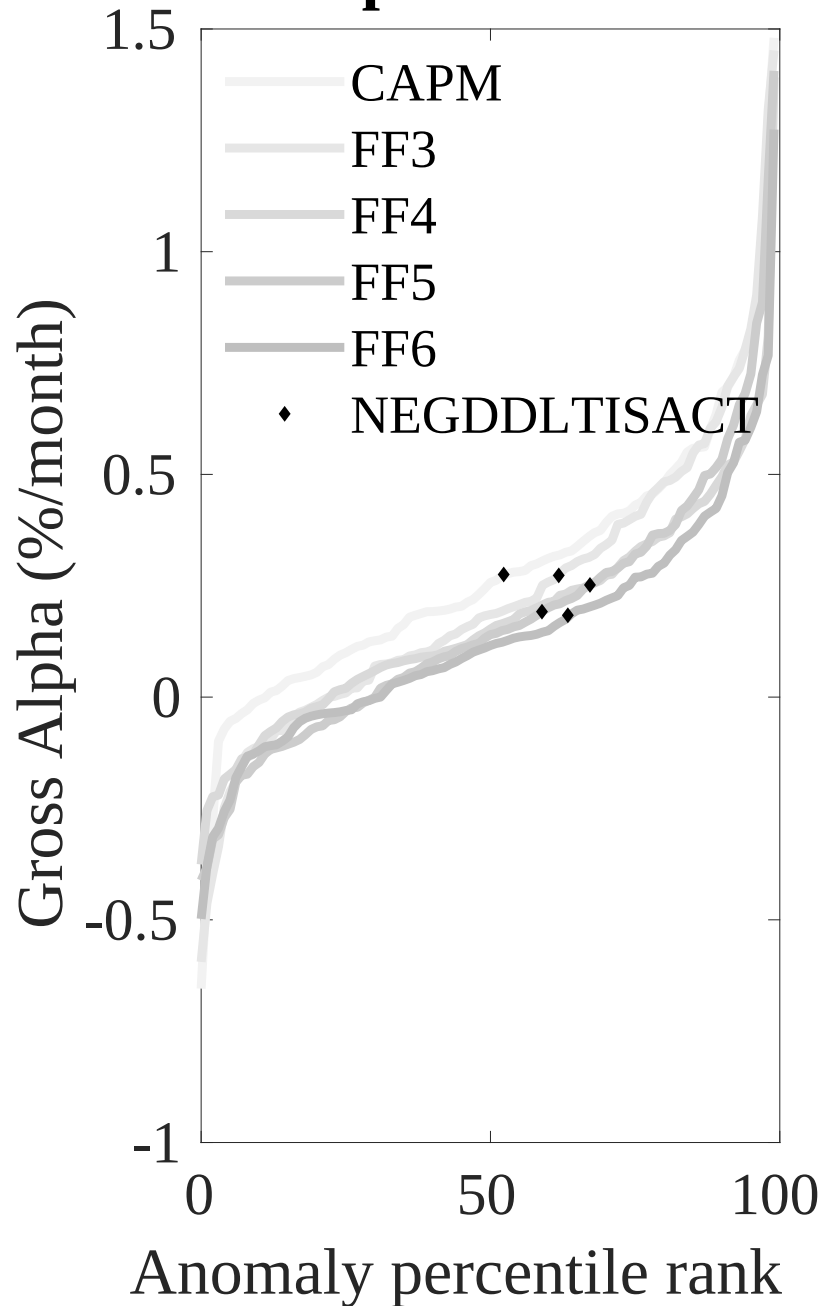
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LLB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

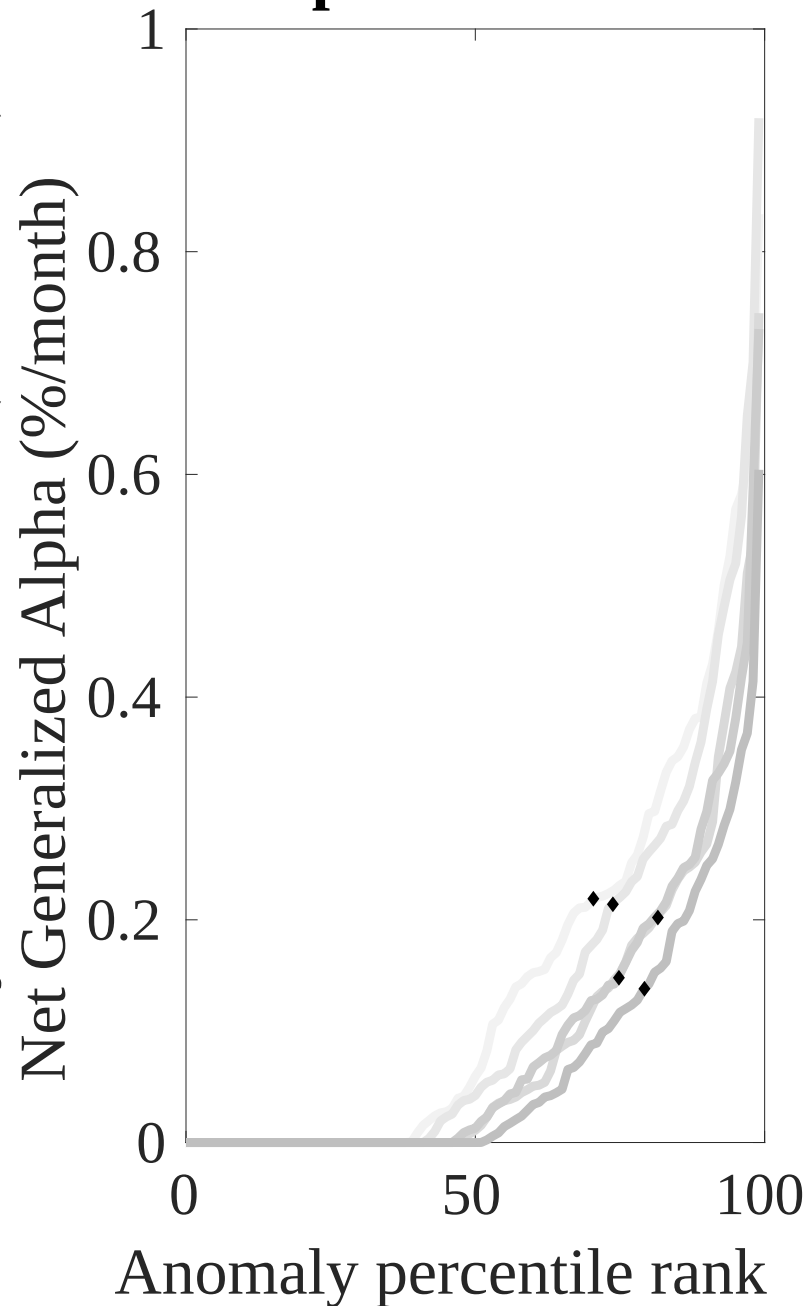
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LLB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

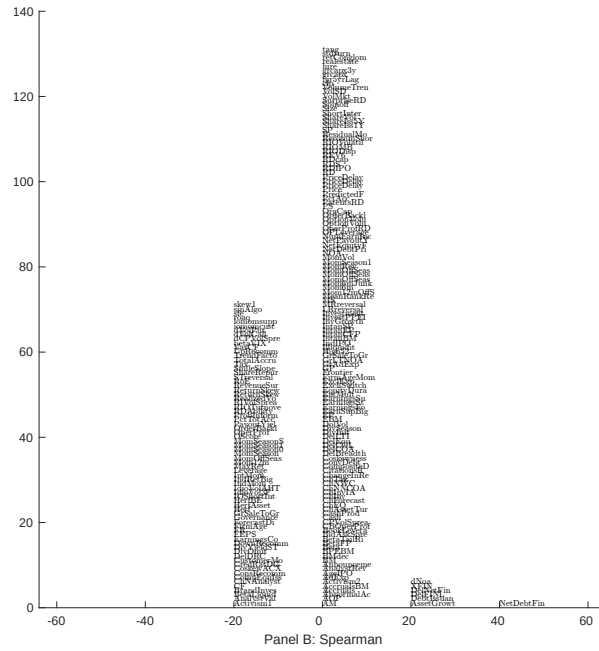
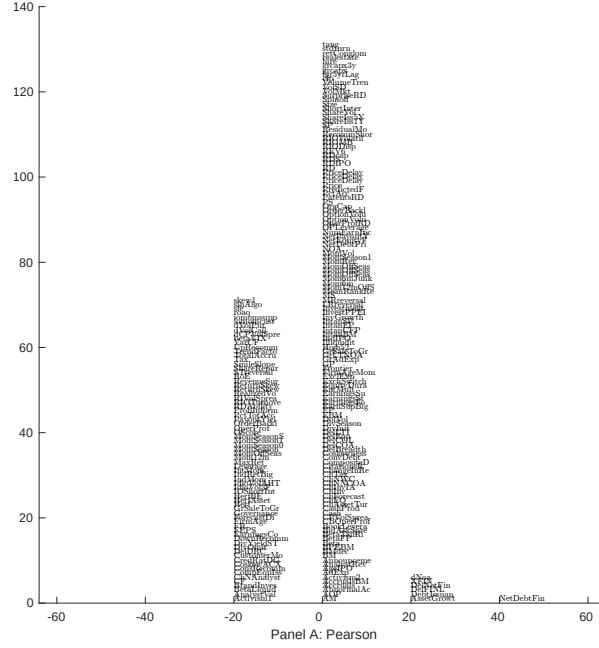
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

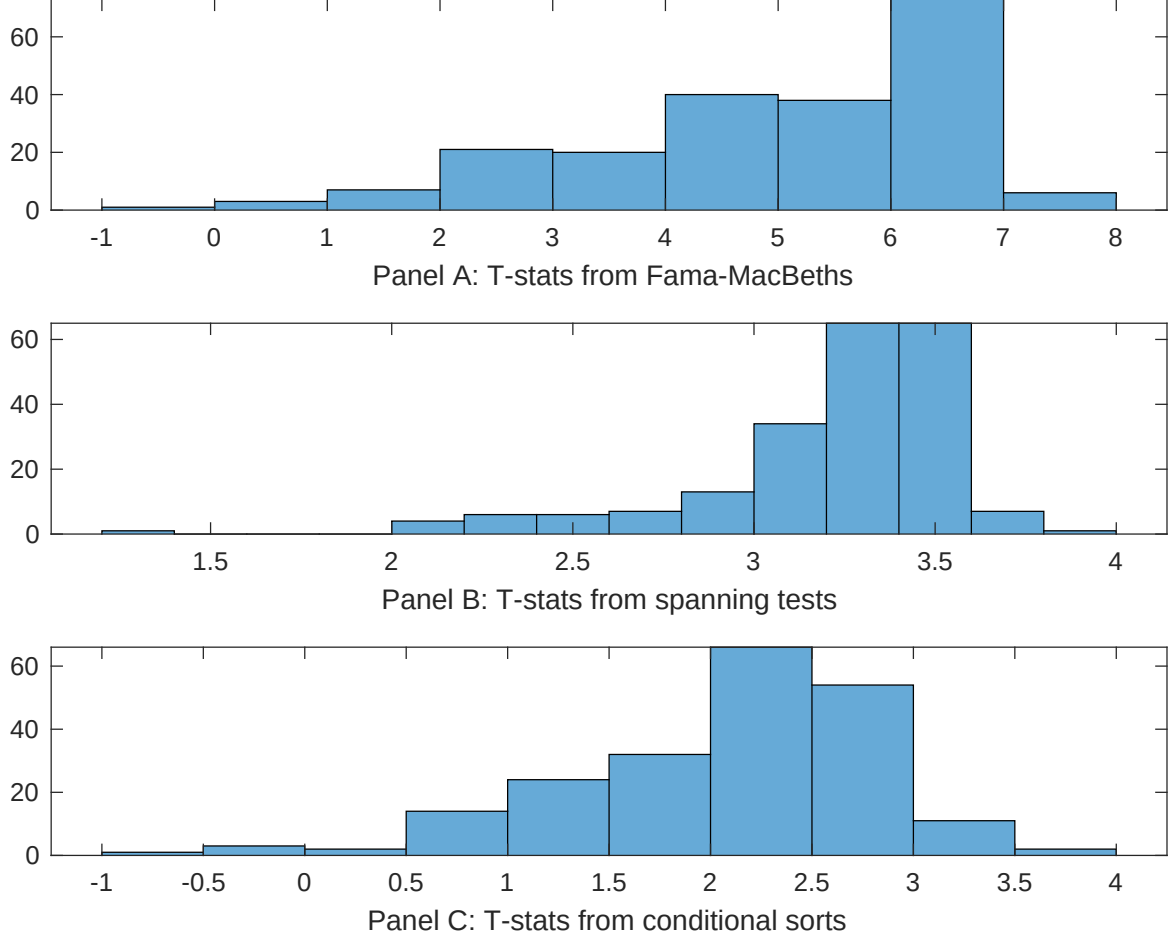
## Net Alpha Percentiles





**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with LLB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LLB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{LLB}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{LLB}LLB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{LLB,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LLB. Stocks are finally grouped into five LLB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLB trading strategies conditioned on each of the 209 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

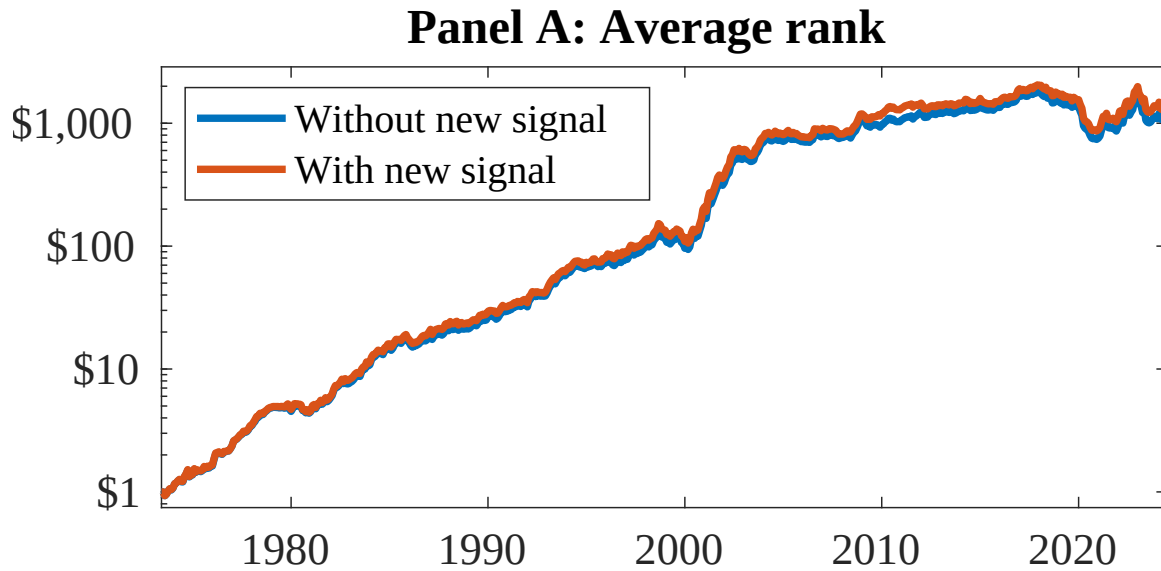
This table presents Fama-MacBeth results of returns on LLB. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{LLB}LLB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.14<br>[5.66] | 0.13<br>[5.24] | 0.14<br>[5.69] | 0.13<br>[5.28] | 0.13<br>[5.30] | 0.14<br>[5.72] | 0.15<br>[5.75]   |
| LLB             | 0.80<br>[2.54] | 0.55<br>[1.85] | 0.53<br>[2.08] | 0.41<br>[1.44] | 0.17<br>[4.84] | 0.28<br>[0.96] | -0.93<br>[-0.25] |
| Anomaly 1       | 0.19<br>[6.23] |                |                |                |                |                | 0.11<br>[1.84]   |
| Anomaly 2       |                | 0.19<br>[8.46] |                |                |                |                | 0.97<br>[1.45]   |
| Anomaly 3       |                |                | 0.17<br>[7.74] |                |                |                | 0.90<br>[3.19]   |
| Anomaly 4       |                |                |                | 0.17<br>[9.17] |                |                | -0.75<br>[-1.57] |
| Anomaly 5       |                |                |                |                | 0.41<br>[6.88] |                | 0.18<br>[0.30]   |
| Anomaly 6       |                |                |                |                |                | 0.11<br>[8.51] | 0.51<br>[2.37]   |
| # months        | 612            | 612            | 612            | 612            | 612            | 612            | 612              |
| $\bar{R}^2(\%)$ | 1              | 0              | 0              | 0              | 0              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LLB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{LLB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

|                 |                  |                  |                   |                  |                  |                  |                  |
|-----------------|------------------|------------------|-------------------|------------------|------------------|------------------|------------------|
| Intercept       | 0.16<br>[2.35]   | 0.17<br>[2.45]   | 0.18<br>[2.60]    | 0.17<br>[2.43]   | 0.18<br>[2.64]   | 0.18<br>[2.56]   | 0.16<br>[2.33]   |
| Anomaly 1       | 13.63<br>[3.97]  |                  |                   |                  |                  |                  | 11.22<br>[2.96]  |
| Anomaly 2       |                  | 12.12<br>[3.22]  |                   |                  |                  |                  | 5.95<br>[1.15]   |
| Anomaly 3       |                  |                  | 7.94<br>[2.55]    |                  |                  |                  | 3.17<br>[0.95]   |
| Anomaly 4       |                  |                  |                   | 9.04<br>[2.31]   |                  |                  | 3.34<br>[0.64]   |
| Anomaly 5       |                  |                  |                   |                  | 6.09<br>[2.26]   |                  | 4.63<br>[1.63]   |
| Anomaly 6       |                  |                  |                   |                  |                  | 4.02<br>[1.00]   | -2.12<br>[-0.51] |
| mkt             | -0.56<br>[-0.34] | -2.41<br>[-1.52] | -2.54<br>[-1.60]  | -2.23<br>[-1.40] | -2.54<br>[-1.60] | -2.44<br>[-1.53] | -0.91<br>[-0.55] |
| smb             | 5.01<br>[1.91]   | -0.44<br>[-0.18] | 0.66<br>[0.27]    | -0.29<br>[-0.12] | 1.30<br>[0.54]   | 0.34<br>[0.14]   | 4.03<br>[1.44]   |
| hml             | -7.64<br>[-2.50] | -8.66<br>[-2.83] | -10.01<br>[-3.25] | -8.55<br>[-2.78] | -9.35<br>[-3.05] | -9.33<br>[-3.03] | -7.87<br>[-2.53] |
| rmw             | -1.76<br>[-0.47] | 5.06<br>[1.62]   | 6.54<br>[2.10]    | 5.58<br>[1.78]   | 7.27<br>[2.32]   | 6.41<br>[2.05]   | -0.55<br>[-0.14] |
| cma             | 15.66<br>[3.05]  | 21.39<br>[4.51]  | 18.26<br>[3.47]   | 21.50<br>[4.45]  | 19.12<br>[3.66]  | 19.66<br>[2.92]  | 10.54<br>[1.51]  |
| umd             | 1.39<br>[0.89]   | 0.70<br>[0.44]   | 1.46<br>[0.93]    | 0.70<br>[0.44]   | 0.86<br>[0.54]   | 1.57<br>[0.99]   | 0.19<br>[0.12]   |
| # months        | 612              | 612              | 612               | 612              | 612              | 612              | 612              |
| $\bar{R}^2(\%)$ | 9                | 8                | 8                 | 8                | 8                | 7                | 10               |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LLB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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